THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two dice are rolled together. The probability that the product of the numbers is 12 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{1}{10}$ **(d)** $\frac{8}{9}$

2. Two dice are rolled together. The probability that the difference of the numbers is 2 is: 1

(a) $\frac{7}{9}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{9}$

3. Three coins are tossed. The probability of getting at least one head is: 1

(a) $\frac{7}{9}$ **(b)** $\frac{7}{8}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{3}{4}$

4. A bag has 6 red balls and some blue balls. If the probability of drawing a blue ball is $\frac{2}{5}$, the number of blue balls is: 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 5

5. A bag has 36 balls, red or blue. If $P(\text{red}) = \frac{5}{12}$, the number of blue balls is: 1

(a) 20 **(b)** 23 **(c)** 21 **(d)** 22

6. A letter is chosen from the word MISSISSIPPI. The probability that it is not the letter I is: 1

(a) $\frac{8}{11}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{7}{11}$

7. The probability that a seed germinates is $\frac{7}{10}$. Out of 2000 seeds the expected number that will NOT germinate is: 1

(a) 600 **(b)** 599 **(c)** 601 **(d)** 602

8. If $P(E) = 0.35$ and $P(F) = 0.45$ and E, F cannot occur together, then $P(E \text{ or } F) =$ 1

(a) 0.10 **(b)** 0.80 **(c)** 0.20 **(d)** 0.1575

9. **Assertion (A):** When a fair die is rolled, the probability of getting a prime number is $\frac{1}{2}$. 1

Reason (R): The prime numbers on a die are 2, 3 and 5, which are 3 of the 6 equally likely outcomes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The probability of an impossible event is 0. 1

Reason (R): The probability of any event lies between 0 and 1, both included.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two dice are rolled together. Find the probability that the product of the numbers is 12. 2

12. A bag has 12 red balls and some blue balls. If $P(\text{blue}) = \frac{1}{4}$, find the number of blue balls. 2

13. Three coins are tossed. Find the probability of getting at least one head, using the complement rule. 2

14. Two dice are rolled together. Find the probability that the product of the numbers is 6. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{3}$ and $P(\text{blue}) = \frac{1}{3}$ and there are 24 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

16. In 100 throws of a die, the number of times a prime number appeared was 52 and a number greater than 4 appeared 31 times (these include 14 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

17. A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is 3
(i) not S,
(ii) a vowel or S.

OR

A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{5}$ and $P(\text{blue}) = \frac{2}{5}$ and there are 40 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

18. In 200 throws of a die, the number of times a prime number appeared was 108 and a number greater than 4 appeared 63 times (these include 30 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

OR

A letter is chosen at random from the word MATHEMATICS. Find the probability that it is 3
(i) not T,
(ii) a vowel or T.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a Ludo game at home, Arjun rolls a fair six-faced die once. 5
(i) How many outcomes are there in the sample space?
(ii) Find the probability of getting an even number.
(iii) Find the probability of getting a number greater than 4.
(iv) Find the probability of getting a multiple of 3 and hence the probability of not getting a multiple of 3.

20. Two dice are rolled together. Find the probability that 5
(i) the sum is 7,
(ii) the sum is more than 7,
(iii) the sum is at most 7. Verify that
(ii) and
(iii) add to 1.

OR

A bag has 9 red, 9 blue and 15 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{4}$. 5
How many green balls were removed?

END OF QUESTION PAPER — Total: 40 Marks